

Plan Sponsor — 401(k) Plan Clinic

A diagnostic dashboard that runs self-service "check-up" tests on a 401(k) plan and reports each one as Pass / Warn / Fail — replacing static quarterly reports with an always-on, AI-assisted clinic for plan health and operations.

Purpose & Value

What it does

Continuously evaluates a plan against compliance, participation, savings, operational, and fiduciary benchmarks; ranks the issues; explains them in plain English; forecasts where each test is heading; and lets the sponsor model fixes before committing to them.

Value delivered

- Replaces 60-page PDFs with a ranked, actionable to-do list
- Catches compliance failures (ADP/ACP/Top-Heavy) before plan year-end
- Quantifies the impact of plan-design changes before adopting them
- Surfaces operational issues (payroll errors, call spikes) in near real-time
- Explains "why" with AI, not just "what" — sponsors don't need to be actuaries

Intended users

401(k) Plan Sponsors	Plan Advisors / TPAs	Recordkeepers
Employers who offer the plan	Consultants supporting sponsors	Diagnostic value-add layer

Functional Modules

Each sidebar tab maps to a module below, with capabilities, intended message, and metric calculations.

Plan Check-Up

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Goal / Message

Give a sponsor a one-glance, doctor-style read on overall plan health and the ranked list of issues that need attention right now.

Capabilities

- Hero Health Score (0–100) with one-line narrative
- Ranked action list — failing tests first, then warn, sorted by impact
- Collapsed neutral list of passing/healthy tests
- Customize sheet to add/remove which tests appear (persists to localStorage)
- Per-test AI Explain button (streamed plain-English) and 12-month forecast mini-chart

Metrics & Calculations

Health Score	Weighted score across all visible tests. Pass = full points, Warn = partial, Fail = 0. Compliance and fiduciary tests carry higher weight than informational ones. Result normalized to 0–100.
Action Ranking	Severity (Fail > Warn > Pass) × Impact (participants affected, \$ at risk, regulatory exposure). Ties broken by deadline proximity.

Plan Health Tests

/plan-health

Goal / Message

Diagnose how well the plan is doing its job for participants — participation, savings behavior, retirement readiness, demographics.

Capabilities

- KPI cards for Participation Rate, Avg Deferral Rate, Auto-Enrollment %, Avg Account Balance
- Demographics toggle (Age vs. Tenure breakdown)
- Click any KPI → 5-year Metric History dialog
- Multi-plan filter (aggregate across Plan IDs)
- Time-range filter: YTD, 1Y, 3Y, 5Y, Since Inception

Metrics & Calculations

Participation Rate	$\text{Active Participants with balance} > \$0 \div \text{Eligible Employees} \times 100$
Average Deferral Rate	$\Sigma(\text{participant deferral \%} \times \text{participant comp}) \div \Sigma(\text{participant comp}) - \text{comp-weighted, not headcount-weighted}$
Auto-Enrollment Rate	$\text{Participants enrolled via auto-enrollment} \div \text{Total participants} \times 100$
Retirement Readiness	$\text{Projected balance at age 65 (current balance compounded at assumed return + future contributions)} \div \text{Target replacement income (typically 80\% of final salary} \times 25)$

Operations Tests

/operations

Goal / Message

Surface operational quality issues — payroll file errors, call center spikes, enrollment activity, compliance test pass/fail.

Capabilities

- Enrollment Activity panel (above Payroll for priority)
- Payroll File Processing — error counts, on-time %, files rejected
- Call Center Volume + interactive bubble cloud of top call reasons
- Advice Requests trend
- Compliance Testing summary (ADP/ACP/Top-Heavy/410(b)/415)

Metrics & Calculations

Payroll On-Time %	Files received by deadline ÷ Total files expected × 100
ADP Test Spread	Avg HCE deferral % – Avg NHCE deferral %. Pass thresholds: ≤ NHCE × 1.25, or ≤ NHCE + 2 (capped at NHCE × 2)
Top-Heavy Ratio	Σ Key Employee account balances ÷ Σ All Employee account balances. > 60% = top-heavy → minimum 3% employer contribution required.
Call Reason Concentration	% of calls in top reason category. Bubble size = call count, color = trend vs prior period.

Recommendations

/recommendations

Goal / Message

Translate diagnostics into specific, prioritized plan-design or operational changes the sponsor can act on.

Capabilities

- Recommendation cards with rationale, expected impact, and effort level
- Linked back to source diagnostic test(s)
- Filter by impact area (participation, cost, compliance, readiness)

Metrics & Calculations

Expected Impact	Modeled delta on target metric (e.g., +X pp participation if auto-enroll default raised from 3% → 6%) using elasticity assumptions from peer-plan data.
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Participant Education (Beta)

/participant-education

Goal / Message

Track education campaigns — reach, engagement, and behavioral lift.

Capabilities

- Campaign history with channel, audience, reach
- Engagement metrics: open rate, click-through, action taken
- Pre/post behavioral lift (deferral rate change, advice utilization)

Metrics & Calculations

Engagement Rate	$\text{Unique participants who took the targeted action} \div \text{Campaign reach} \times 100$
Behavioral Lift	$(\text{Post-campaign metric} - \text{Pre-campaign metric}) \div \text{Pre-campaign metric} \times 100$

Documents (Beta)

/documents

Goal / Message

Centralized repository of plan filings (Form 5500, SPD, amendments, audits) with filtering and downloads.

Capabilities

- Filter by year, document type, status
- One-click download
- Filing-deadline indicators

Metrics & Calculations

Document Currency	$\text{Documents updated within required cycle} \div \text{Total required documents} \times 100$
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What-If Simulator (AI)

/what-if

Goal / Message

Let the sponsor type a plan-design scenario in plain English and see modeled impact across the metrics that matter.

Capabilities

- Free-text prompt + example chips
- AI returns structured impact across Participation, Avg Deferral, ADP Spread, Total Annual Plan Cost
- Each metric shows current → projected → delta → rationale
- Assumptions and recommendation summary

Metrics & Calculations

Projected Participation	Current participation + elasticity coefficient × magnitude of design change (e.g., raising auto-enroll default by 1pp ≈ +0.4pp participation; adding match ≈ +5-12pp depending on richness)
Projected Annual Plan Cost	$\Sigma(\text{employer match per participant}) + \text{admin fees}$. Match = $\min(\text{deferral \%}, \text{match cap}) \times \text{match rate} \times \text{eligible comp}$.

RetirementAI (Beta)

/retirementiq

Goal / Message

Conversational AI assistant for plan questions, answered with plan context.

Capabilities

- Streamed chat interface (Gemini via edge function)
- Maintains conversation history
- Cites the metric or test that informed each answer

Metrics & Calculations

Response Grounding	Answers are constrained to the plan's loaded metrics and diagnostic results – no free-form numerical claims.
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Usage (Beta)

/usage

Goal / Message

Internal view of platform activity and AI cost.

Capabilities

- Active users, sessions, feature adoption
- AI calls by feature, token usage, cost

Metrics & Calculations

Cost per Active User	$\text{Total AI + infra cost in period} \div \text{Distinct active users in period}$
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Data Schema

/data-schema

Goal / Message

Reference of every table, column, and data type required to power the dashboard.

Capabilities

- 21 tables grouped by domain (Plan, Participant, Contributions, Tests, Operations, AI)
- Column names, data types, primary/foreign keys

Reference Architecture

Current Lovable build maps cleanly to a production GCP deployment.

Frontend	React 18 + Vite + TypeScript, Tailwind + shadcn/ui, React Router, TanStack Query, Recharts. Static SPA bundle.
Backend (target: GCP)	FastAPI on Cloud Run, Cloud SQL (PostgreSQL, 21 tables), Vertex AI / Gemini for AI features, Firebase Auth, Firebase Hosting for the SPA, Artifact Registry for images.

See the Data Schema tab for the full table/column inventory, and the project tickets (PC-001 → deployment) for the build sequence.